Q1 - 24 June - Shift 2

Let the maximum area of the triangle that can be inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{4}=1$, $a > 2$, having one of its vertices at one end of the major axis of the ellipse and one of its sides parallel to the y-axis, be $6\sqrt{3}$. Then the eccentricity of the ellipse is :

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{4}$

Space for your notes:

Q2 - 25 June - Shift 2

The line $y=x+1$ meets the ellipse $\frac{x^2}{4}+\frac{y^2}{2}=1$ at two points P and Q. If r is the radius of the circle with PQ as diameter then $(3r)^2$ is equal to

(A) 20
(B) 12
(C) 11
(D) 8

Space for your notes:

Q3 - 26 June - Shift 1

Let the common tangents to the curves $4(x^2 + y^2) = 9$ and $y^2 = 4x$ intersect at the point Q. Let an ellipse, centered at the origin O, has lengths of semi-minor and semi-major axes equal to OQ and 6, respectively. If e and l respectively denote the eccentricity and the length of the latus rectum of this ellipse, then $\frac{l}{e^2}$ is equal to____________.

Space for your notes:

Q4 - 26 June - Shift 2

If m is the slope of a common tangent to the curves $\frac{x^2}{16}+\frac{y^2}{9}=1$ and $x^2+y^2=12$, then $12m^2$ is equal to :

(A) 6 (B) 9

(C) 10 (D) 12

Space for your notes:

Q5 - 26 June - Shift 2

The locus of the mid point of the line segment joining the point (4, 3) and the points on the ellipse $x^2+2y^2=4$ is an ellipse with eccentricity :

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2\sqrt{2}}$

(C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

Space for your notes:

Q6 - 27 June - Shift 1

Let the eccentricity of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,\ a>b$, be $\frac{1}{4}$. If this ellipse passes through the point $\left(-4\sqrt{\frac{2}{5}},3\right)$, then a^2+b^2 is equal to :

(A) 29 (B) 31

(C) 32 (D) 34

Space for your notes:

Q7 - 29 June - Shift 1

Let PQ be a focal chord of the parabola $y^2 = 4x$ such that it subtends an angle of $\frac{\pi}{2}$ at the point (3, 0). Let the line segment PQ be also a focal chord of the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a^2 > b^2$. If e is the eccentricity of the ellipse E, then the value of $\frac{1}{e^2}$ is equal to :

(A) $1 + \sqrt{2}$ (B) $3 + 2\sqrt{2}$

(C) $1 + 2\sqrt{3}$ (D) $4 + 5\sqrt{3}$

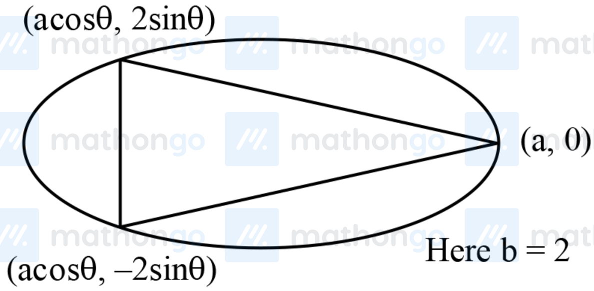
Space for your notes:

Answer Key

Q1 (A) **Q2** (A) **Q3** (4) **Q4** (B)

Q5 (C) **Q6** (B) **Q7** (B)

Q1 (A)



$A = \frac{1}{2} a(1 - \cos\theta)(4\sin\theta)$

$A = 2a(1 - \cos\theta)\sin\theta$

$\frac{dA}{d\theta} = 2a\left(\sin^2\theta + \cos\theta - \cos^2\theta\right)$

$\frac{dA}{d\theta} = 0 \Rightarrow 1 + \cos\theta - 2\cos^2\theta = 0$

$\cos\theta = 1$ (Reject)

OR

$\cos\theta = \frac{-1}{2} \Rightarrow \theta = \frac{2\pi}{3}$

$\frac{d^2A}{d\theta^2} = 2a\left(2\sin^2\theta - \sin\theta\right)$

$\frac{d^2A}{d\theta^2} < 0$ for $\theta = \frac{2\pi}{3}$

Now, $A_{max} = \frac{3\sqrt{3}}{2} a = 6\sqrt{3}$

$\boxed{a = 4}$

Now, $e = \sqrt{\frac{a^2 - b^2}{a^2}} = \frac{\sqrt{3}}{2}$ **Ans. (A)**

Q2 (A)

Ellipse $x^2 + 2y^2 = 4$

Line $y = x + 1$

Point of intersection

$$x^2 + 2(x+1)^2 = 4$$

$$3x^2 + 4x - 2 = 0$$

$$|x_1 - x_2| = \frac{\sqrt{40}}{3}$$

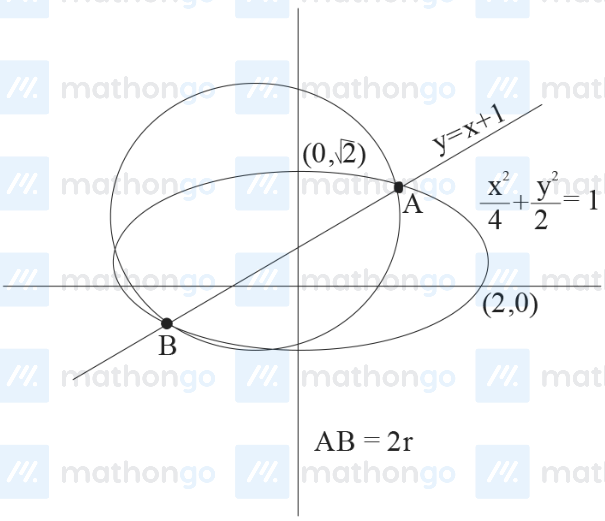
$$AB = 2r = |x_1 - x_2|\sqrt{1+m^2},$$

m is slope of given line

$$AB = \frac{\sqrt{40}}{3}\sqrt{1+1}$$

$$2r = \frac{\sqrt{80}}{3} \Rightarrow r = \frac{\sqrt{80}}{6}$$

$$(3r)^2 = \left(3 \times \frac{\sqrt{80}}{6}\right)^2 = \frac{80}{4} = 20$$



Q3 (4)

$$x^2 + y^2 = \frac{9}{4} \qquad y = 4x$$

Equation tangent in slope form

$y = mx \pm \frac{3}{2}\sqrt{(1+m^2)} \quad ...(1)$

$y = mx + \frac{1}{m} \quad ...(2)$

compare (1) & (2)

$\pm\frac{3}{2}\sqrt{(1+m^2)} = \frac{1}{m^2}$

$9m^2(1 + m^2) = 4$

$9m^4 + 9m^2 - 4 = 0$

$9m^4 + 12m^2 - 3m^2 - 4 = 0$

$3m^2(3m^2 + 4) - (3m^2 + 4) = 0$

$m^2 = -\frac{4}{3}$ (Rejected)

$m^2 = \frac{1}{3} \Rightarrow m = \pm\frac{1}{\sqrt{3}}$

Equation of common tangent

$y = \frac{1}{\sqrt{3}}x + \sqrt{3}$

on X axis y = 0

OQ = –3

b = |OQ| = 3

a = 6

$b^2 = a^2(1 - e^2) \Rightarrow e^2 = 1 - \frac{9}{36} = \frac{3}{4}$

$e = \frac{2b^2}{a} = \frac{2\times 9}{6} = 3$

$\frac{e}{e^2} = \frac{3}{3/4} = 4$

Q4 (B)

$$\frac{x^2}{16}+\frac{y^2}{9}=1$$

equation of tangent to the ellipse is

$$y=mx\pm\sqrt{a^2m^2+b^2}$$

$$y=mx\pm\sqrt{16m^2+9} \quad \ldots\text{(i)}$$

$$x^2+y^2=12$$

equation of tangent to the circle is

$$y=mx\pm\sqrt{12}\sqrt{1+m^2} \quad \ldots\text{(ii)}$$

for common tangent equate eq. (i) and (ii)

$$\Rightarrow 16m^2+9=12(1+m^2)$$

$$16\,m^2-12\,m^2=3$$

$$4\,m^2=3$$

$$12\,m^2=9$$

Q5 (C)

$$\frac{x^2}{4}+\frac{y^2}{2}=1$$

P(4,3) —— D —— Q $\left(2\cos\theta,\sqrt{2}\sin\theta\right)$

Coordinate of D is

$$\left(\frac{2\cos\theta+4}{2},\frac{\sqrt{2}\sin\theta+3}{2}\right)\equiv(h,k)$$

$$\frac{2h-4}{2}=\cos\theta \quad \ldots\text{(i)}$$

$$\frac{2k-3}{\sqrt{2}}=\sin\theta \quad \ldots\text{(ii)}$$

$\text{(i)}^2+\text{(ii)}^2$, then we get

$$\left(\frac{2h-4}{2}\right)^2+\left(\frac{2k-3}{\sqrt{2}}\right)^2=1 \Rightarrow \frac{(x-2)^2}{1}+\frac{\left(y-\frac{3}{2}\right)^2}{\left(\frac{1}{2}\right)}=1$$

$\therefore$ Required eccentricity is

$$e=\sqrt{1-\frac{1}{2}}=\frac{1}{\sqrt{2}}$$

Q6 (B)

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad a>b$$

$$e^2=1-\frac{b^2}{a^2}$$

$$\frac{1}{16}=1-\frac{b^2}{a^2}$$

$$\frac{b^2}{a^2}=1-\frac{1}{16}=\frac{15}{16} \Rightarrow b^2=\frac{15}{16}a^2$$

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

$$\frac{16\times\frac{2}{5}}{a^2}+\frac{9}{b^2}=1$$

$$\frac{32}{5a^2}+\frac{9}{b^2}=1$$

$$\frac{32}{5a^2}+\frac{9}{\frac{15}{16}a^2}=1$$

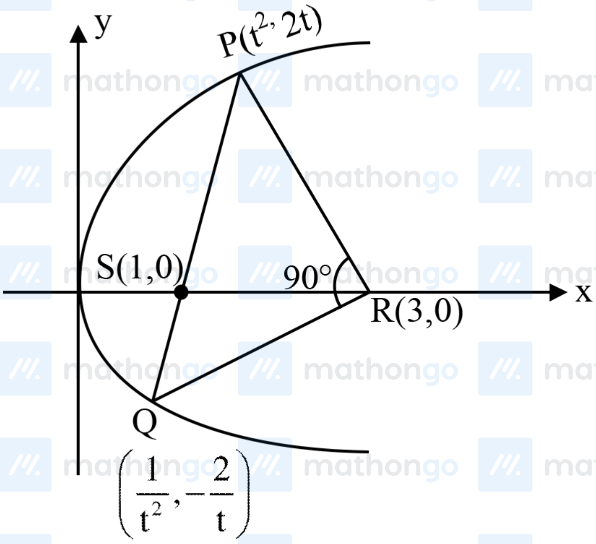
$$\frac{80}{5a^2}=1$$

$$16=a^2$$

$$b^2=15$$

Q7 (B)

PQ is focal chord



$m_{PR} \cdot m_{PQ} = -1$

$$\frac{2t}{t^2-3} \times \frac{-2/t}{\frac{1}{t^2}-3} = -1$$

$(t^2 - 1)^2 = 0$

$\Rightarrow t = 1$

$\Rightarrow$ P & Q must be end point of latus rectum:

P(1, 2) & Q(1, –2)

$\therefore \frac{2b^2}{a} = 4$ & $ae = 1$

$\because$ We know that $b^2 = a^2(1 - e^2)$

$\therefore a = 1 + \sqrt{2}$

$\because e^2 = 1 - \frac{b^2}{a^2}$

$\therefore e^2 = 3 - 2\sqrt{2}$

$\frac{1}{e^2} = 3 + 2\sqrt{2}$